

Introduction to Computer Science (CS10070902)

Your class ID: CS10070902

DO NOT ENTER THE WRONG CLASSROOM!

- **Myself**

- <http://cs1.tongji.edu.cn/~junqiao/>

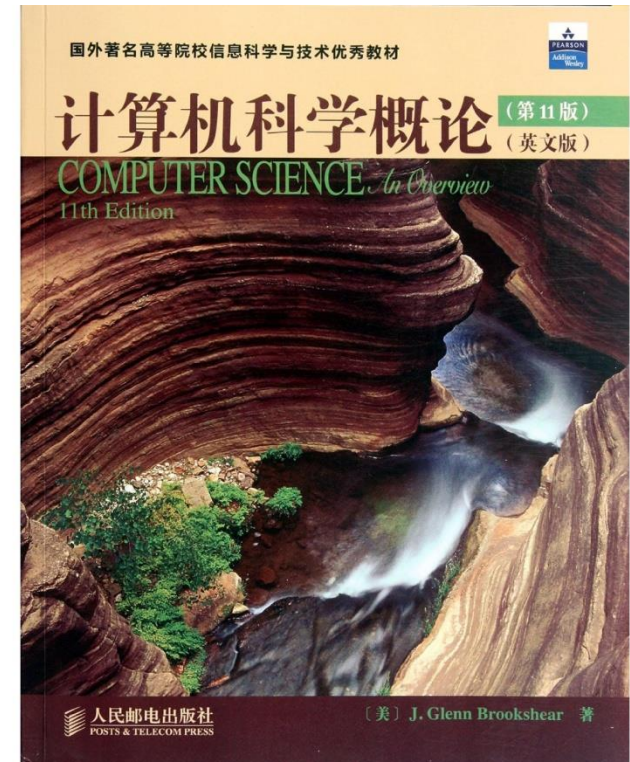
Goal of CS10070902

- Systematic Overview of Computer Science
- Interests and Confidence
- Collaboration and A Taste of Frontiers



Textbook and course website

- **Textbook:**
Computer Science: An Overview
(11th ed) by J. Glenn Brookshear
- Course website :
 - **canvas**
 - Legacy:
<http://cs1.tongji.edu.cn/courses/CS100383>
- Discussion group
 - **QQ 617462497**



教学进度安排

Scoring

- Final Score = Projects (30%) + Attendance (10%) + Online Quiz (15%) + Exam (45%)

Projects

- Detailed descriptions will be released on canvas
 - I Web programming with FLASK
 - II A Glimpse of Computer Science
- About grouping
 - 4 ~ 6 persons per group
- Scoring

- **The general idea of
Computer Science**

- **Information**

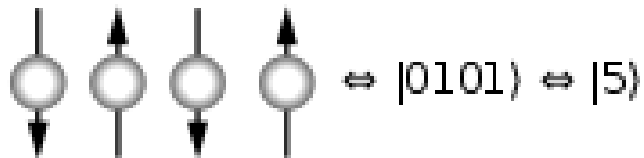
- 1 0
- Bit
- Why binary digit?



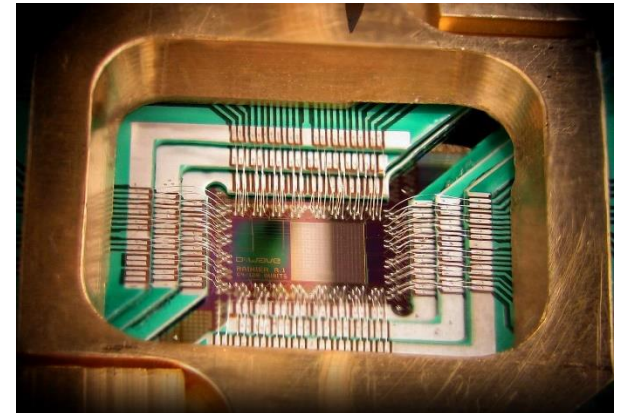
Credit: Koldunov iStock Getty Images

Quantum computer

- Bit $\rightarrow$ Qubit



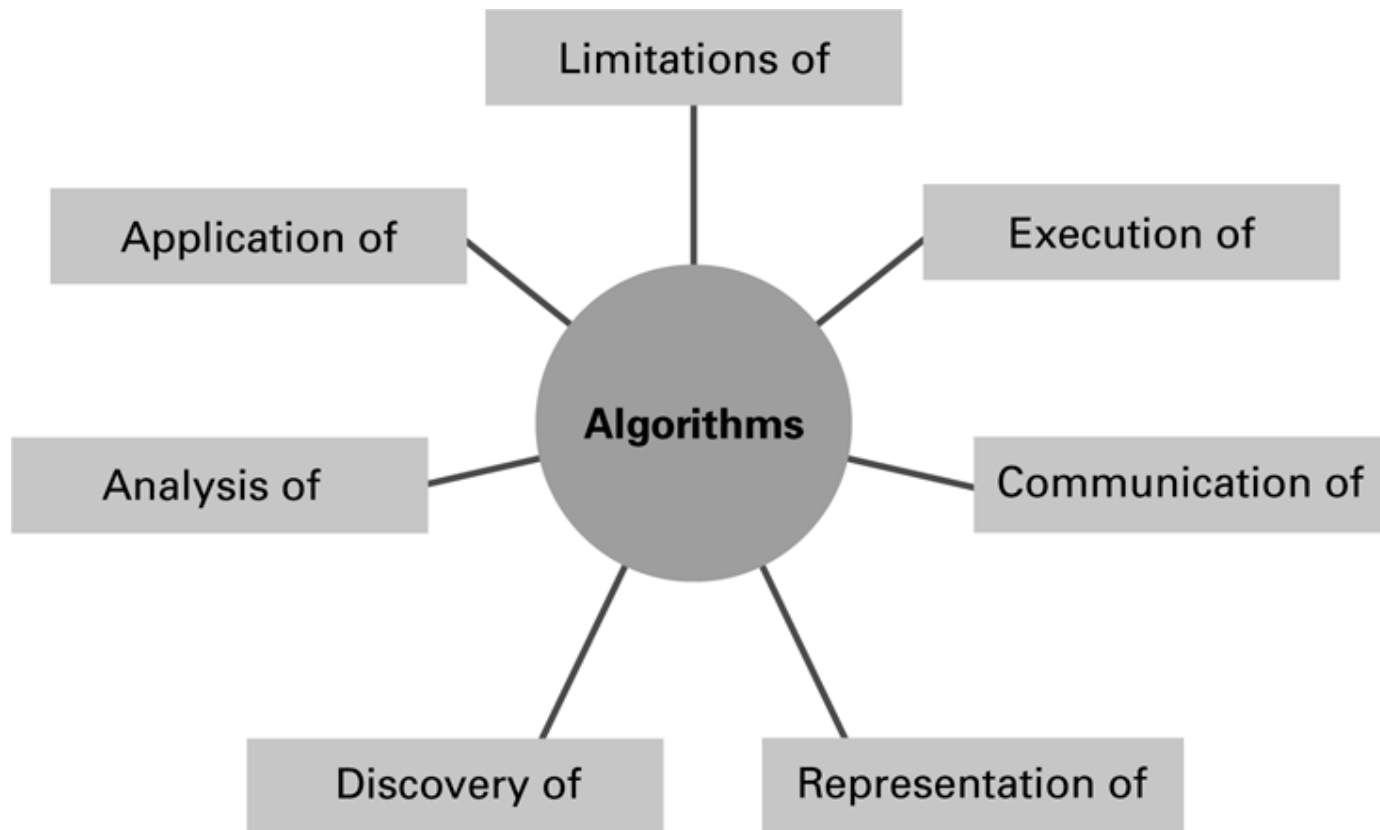
qubits can be in a superposition of all the classically allowed states



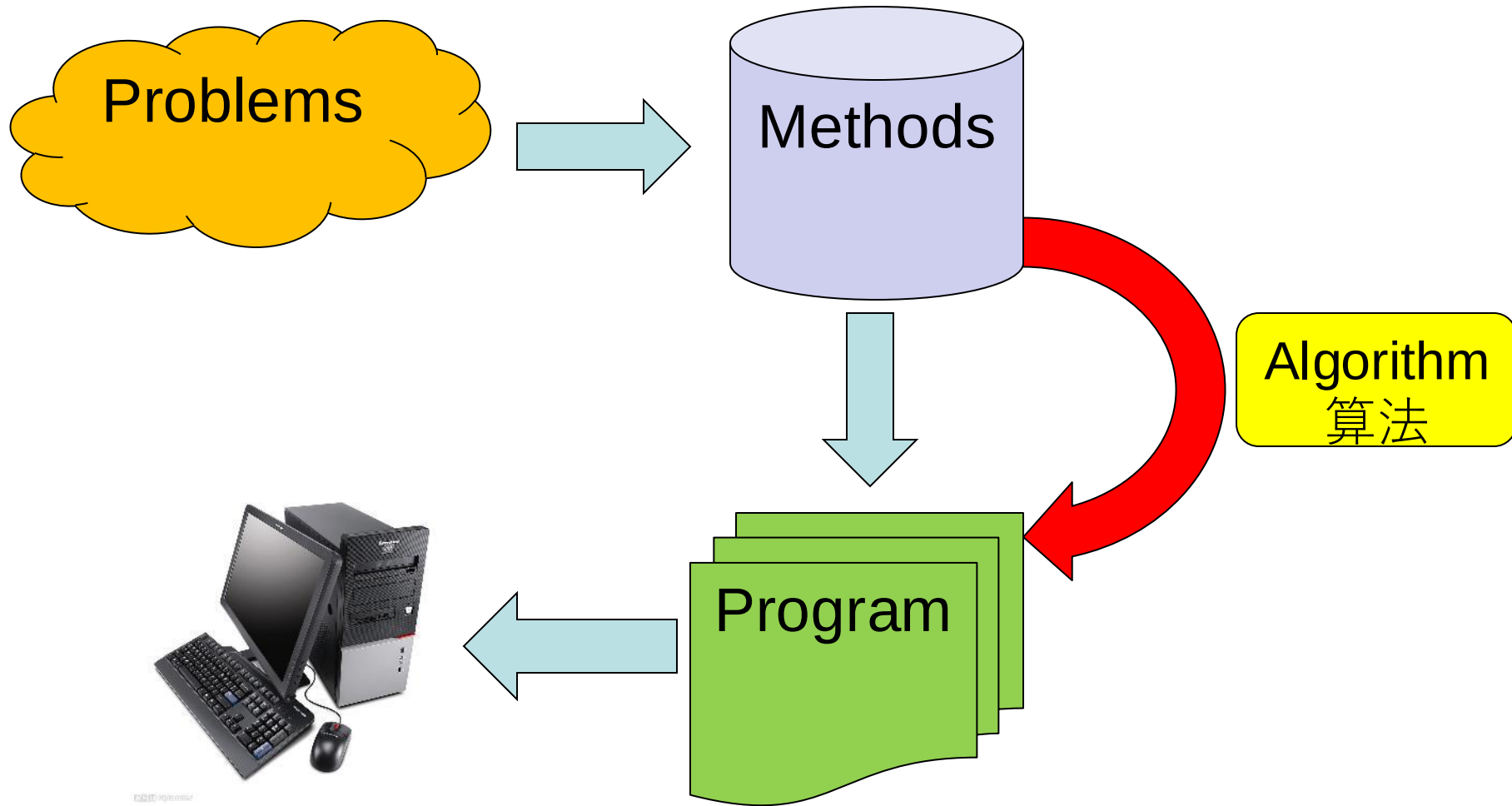
D-Wave

- **Algorithm**
- *The heart of CS*

Central role of algorithms



Program



The first lady

- Ada Lovelace
- 1843



Diagram for the computation by the Engine of the Numbers of Bernoulli. See Note G. (page 722 *et seq.*)

Number of Operation.	Nature of Operation.	Variables acted upon.	Variables receiving results.	Indication of change in the value on any Variable.	Statement of Results.	Data.												Working Variables.												Result Variables.			
						v_1	v_2	v_3	v_4	v_5	v_6	v_7	v_8	v_9	v_{10}	v_{11}	v_{12}	v_{13}	v_{14}	v_{15}	v_{16}	v_{17}	v_{18}	v_{19}	v_{20}	v_{21}	v_{22}	v_{23}	v_{24}	v_{25}	v_{26}		
						0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
						1	2	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
1	$\times$	$v_1 \times v_4$	v_4	v_4	$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= 2n$	2	n	2n	2n	2n																						
2	$-$	$v_4 - v_1$	v_4		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= 2n - 1$	1	...	2n - 1																								
3	$+$	$v_4 + v_1$	v_4		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= 2n + 1$	1	...	2n + 1																								
4	$+$	$v_4 + v_4$	v_{11}		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$\frac{2n - 1}{2}$	...	...	0	0																							
5	$+$	$v_{11} + v_4$	v_{11}		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$\frac{2n - 1}{2}$	2	...	$\frac{2n - 1}{2}$																								
6	$-$	$v_{11} - v_4$	v_{10}		$v_{11} = v_{11}$ $v_{11} = v_{11}$ $v_{11} = v_{11}$	$= \frac{1}{2} \frac{2n - 1}{2n + 1} = A_0$	...	...																									
7	$-$	$v_4 - v_1$	v_{10}		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= n - 1 (= 3)$	1	...	n																								
8	$+$	$v_4 + v_1$	v_7		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= 2 + 0 = 2$	2	...																									
9	$+$	$v_4 + v_7$	v_{11}		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= \frac{2n}{2} = A_1$	...	...																									
10	$\times$	$v_{11} \times v_7$	v_{12}		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= B_1, \frac{2n}{2} = B_1 A_1$	...	...																									
11	$+$	$v_{12} + v_7$	v_{13}		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= \frac{1}{2} \frac{2n - 1}{2n + 1} + B_1, \frac{2n}{2}$	...	...																									
12	$-$	$v_{10} - v_1$	v_{16}		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= n - 2 (= 2)$	1	...																									
13	$-$	$v_4 - v_1$	v_6		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= 2n - 1$	1	...																									
14	$+$	$v_4 + v_7$	v_7		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= 2 + 1 = 3$	1	...																									
15	$+$	$v_4 + v_7$	v_8		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= \frac{2n - 1}{3}$	...	...																									
16	$\times$	$v_4 \times v_7$	v_{11}		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= \frac{2n}{2} \frac{2n - 1}{3}$	...	...																									
17	$-$	$v_4 - v_1$	v_6		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= 2n - 2$	1	...																									
18	$+$	$v_4 + v_7$	v_7		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= 3 + 1 = 4$	1	...																									
19	$+$	$v_4 + v_7$	v_9		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= \frac{2n - 2}{4}$	...	...																									
20	$\times$	$v_4 \times v_7$	v_{11}		$v_4 = v_4$ $v_4 = v_4$ $v_4 = v_4$	$= \frac{2n - 2}{4} \frac{2n - 2}{3} = A_3$	...	...																									

In the new era of machine learning?

- Does human devise Algorithms?
 - **Neural Network**
 - **AutoML**
 - **Reinforcement Learning**
 - **etc**

What is computer science?

- **Computer science** is a discipline that deals with the *theoretical foundations of information and computation*, together with *practical techniques for the implementation and application of these foundations*

- Computer science is about Computational Thinking
 - Abstraction
 - Automation
 - Analysis
 - Autonomous

Abstraction

- **Abstraction:** The distinction between the external properties of an entity and the details of the entity's internal composition
- **Abstract tool:** A “component” that can be used without concern for the component's internal properties

Terminology

- **Algorithm:** A set of steps that defines how a task is performed
- **Program:** A representation of an algorithm
- **Programming:** The process of developing a program
- **Software:** Programs and algorithms
- **Hardware:** Equipment

What Are We Going to Learn?

- Part I: How computers work?
 - Computer hardware
 - Data representation
 - Data manipulation
 - Operating system
 - Computer network

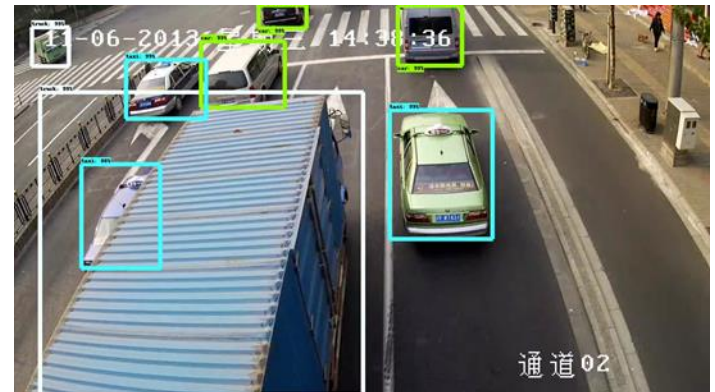
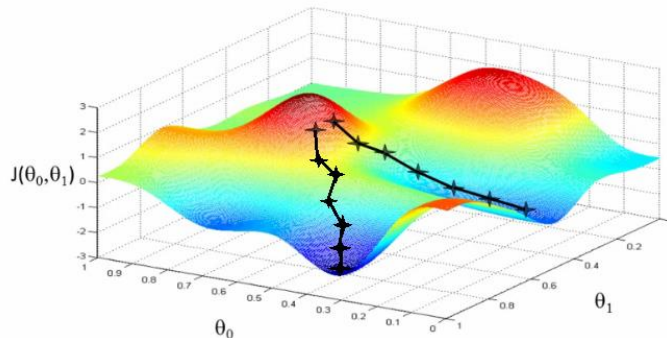
What Else to Learn?

- Part II: How to make computers work?
 - Algorithm
 - Programming
 - Software Engineering



What Else to Learn?

- Part III: What do computers work for?
 - AI
 - Database
 - Computer Graphics

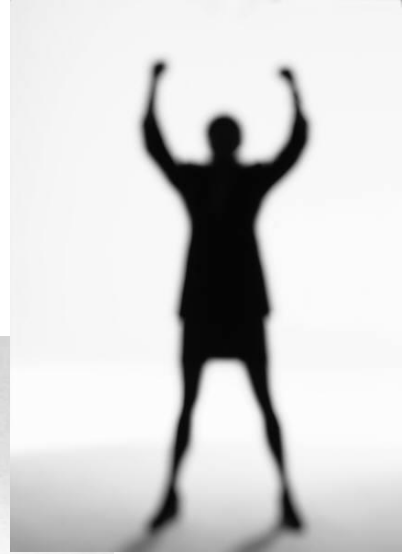
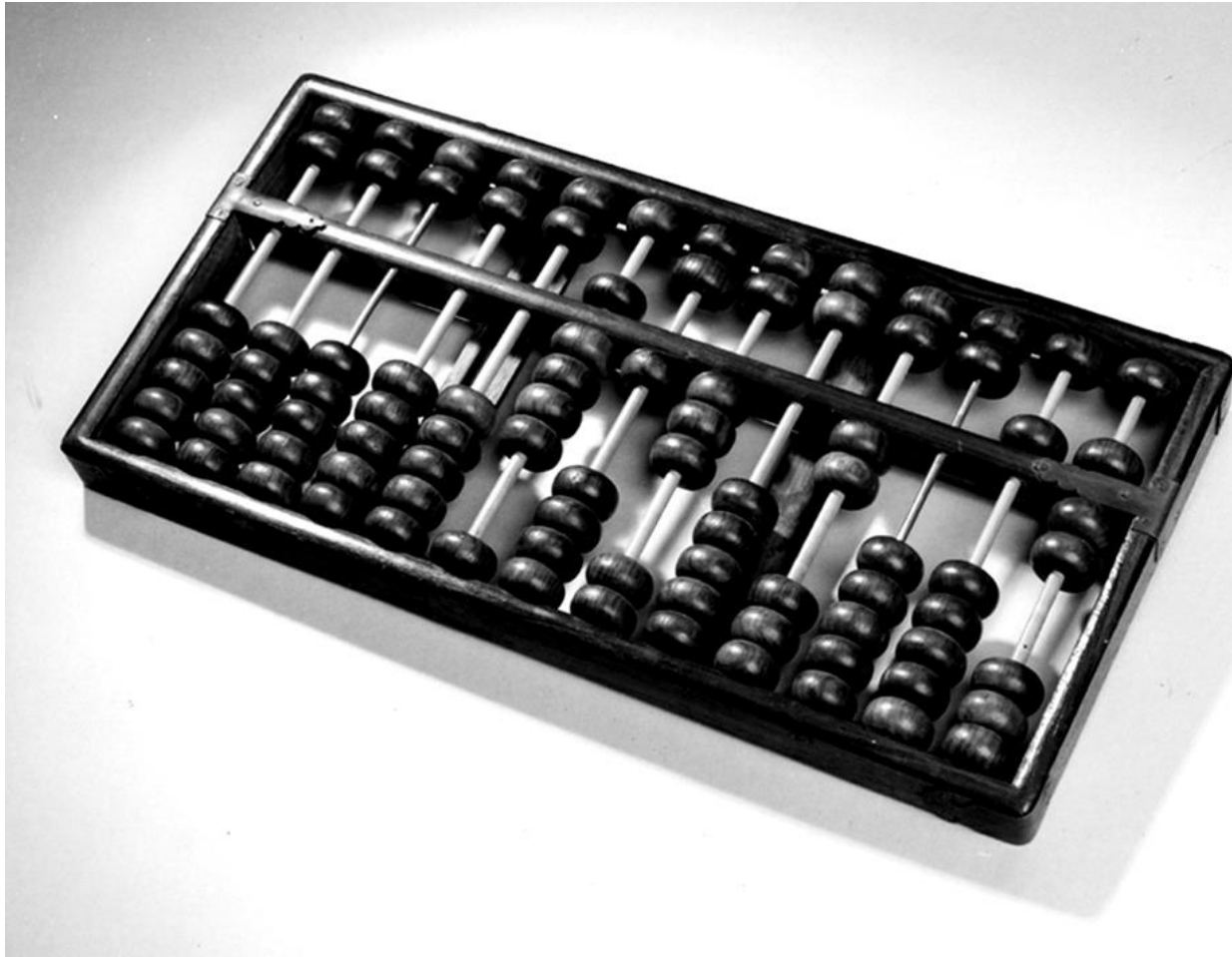


- Questions?

History

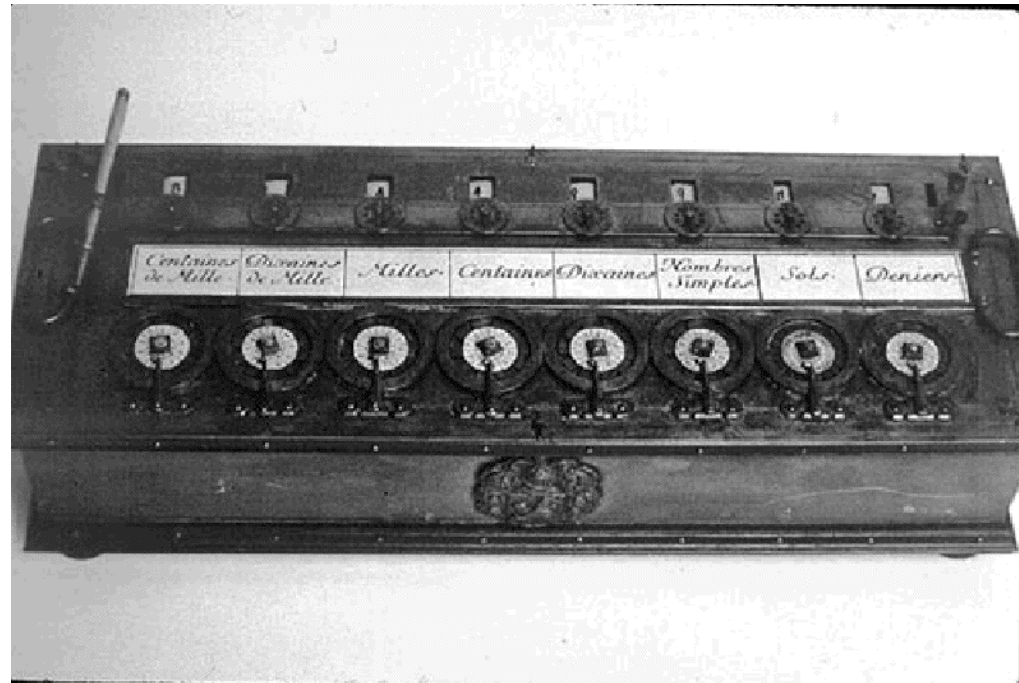
- Early computing devices
 - Abacus: positions of beads represent numbers
 - Gear-based machines (1600s-1800s)
 - Positions of gears represent numbers
 - Blaise Pascal, Wilhelm Leibniz, Charles Babbage

An Abacus



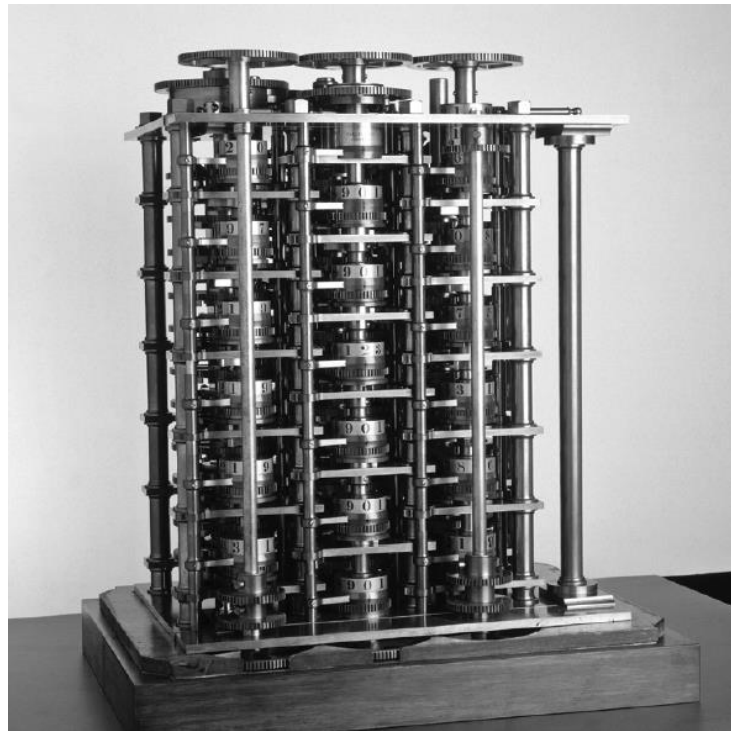
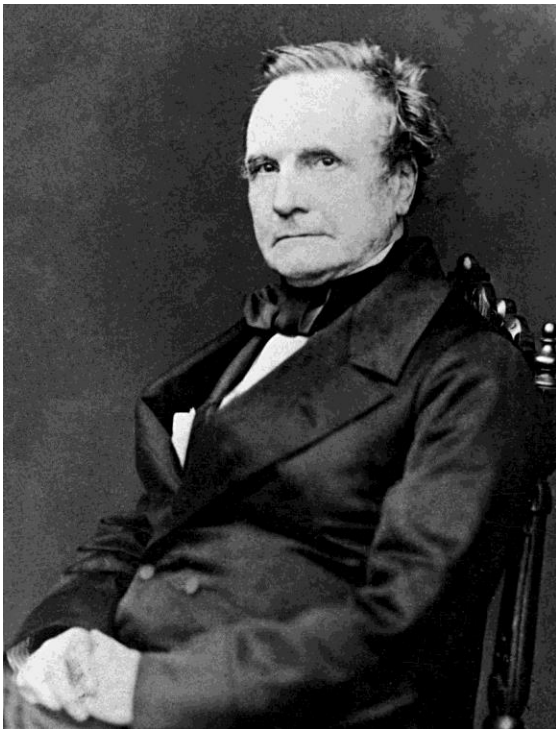
Pascal's Calculating Machine

- Performs basic arithmetic operations (early to mid 1600's). Does not have what may be considered the basic parts of a computer.



Babbage's Difference Engine #1

- Working portion of Babbage's Difference Engine No. 1, which is the first known automatic calculator.

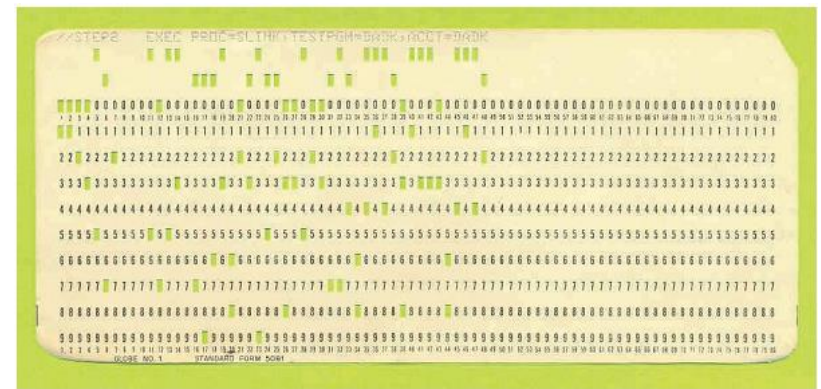


- Video

The Imitation Game



Early Data Storage



- Punched cards
 - First used in Jacquard Loom (1801) to store patterns for weaving cloth
 - Storage of programs in Babbage's Analytical Engine (ADA)
 - Popular through the 1970's
- Gear positions



The Mark I computer



ENIAC (Electronic Numerical Integrator And Calculator)





Alan Turing

Turing Award, is an annual award given
by the Association for Computing
Machinery (ACM) 1966

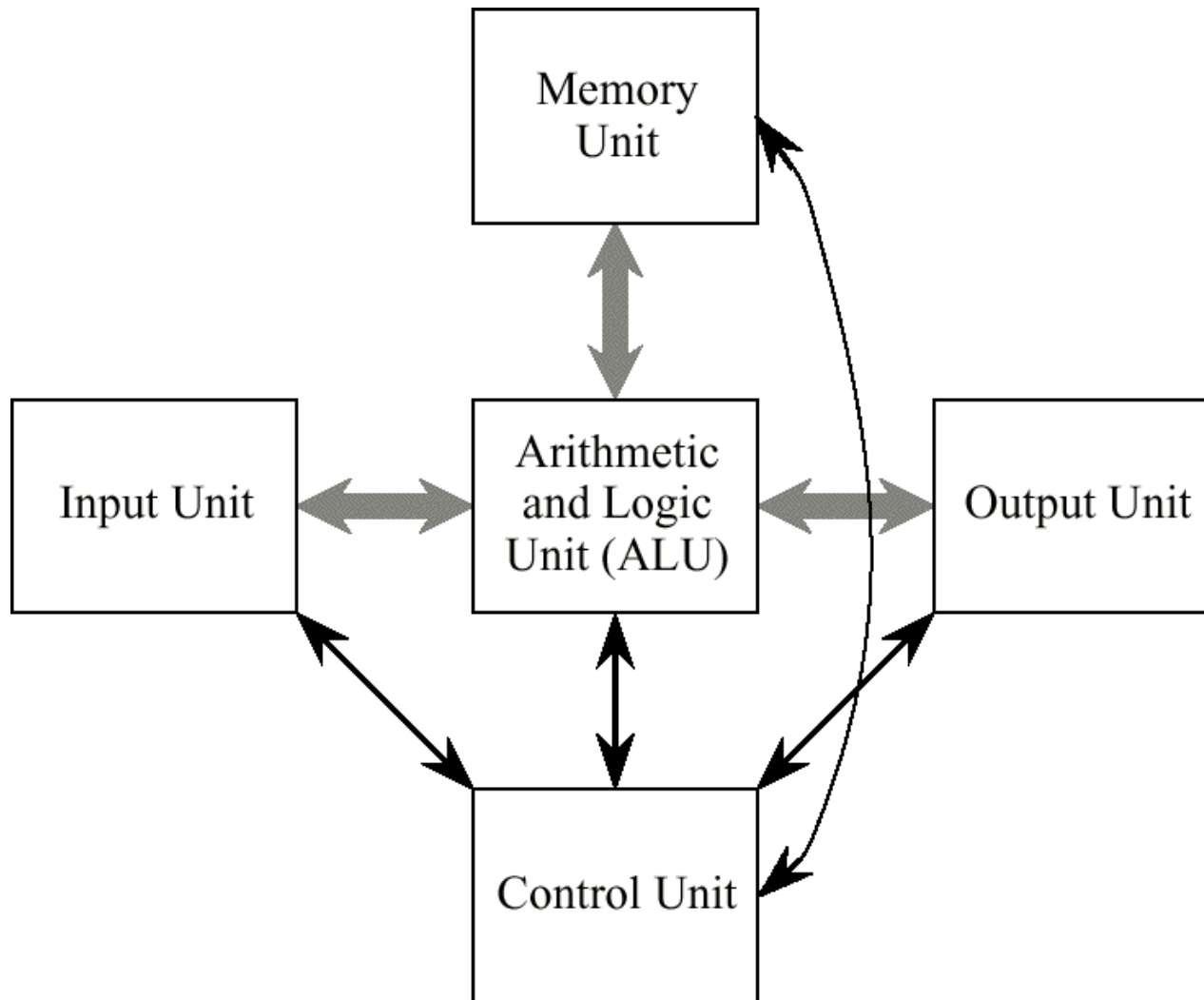


John Von Neumann

Inventor of **stored program**
computers

- A Turing machine

Von Neumann Model



Personal Computers

- First used by hobbyists 1970s
 - Xerox Alto
 - IBM
 - Apple I II
- IBM introduced “the PC” in 1981.
 - Accepted by business
 - Became the standard hardware design for most desktop computers
 - Most PCs use software from Microsoft

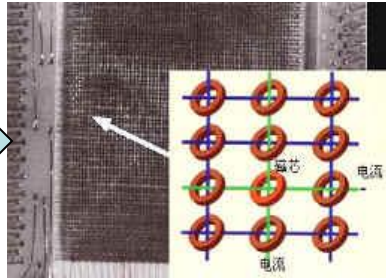
Generations: 1-4

真空管



Vacuum Tube
Computer

晶体管



Transistorized
Computers

集成电路



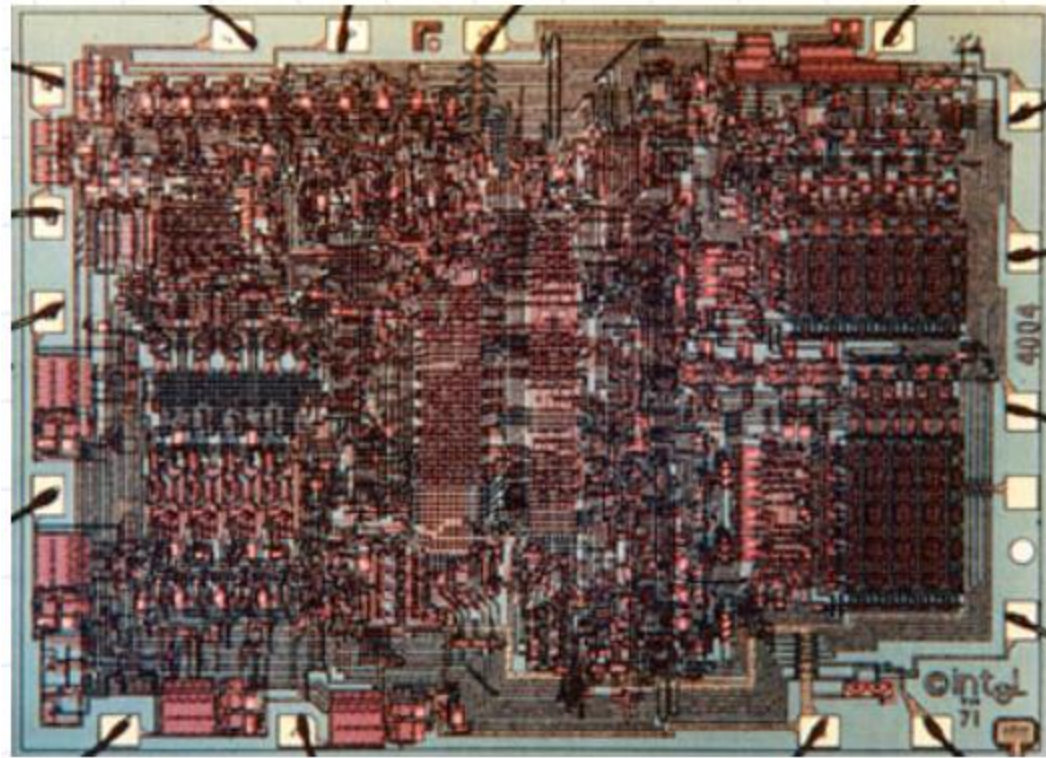
Integrated
Circuit
Computers

大规模集成电路



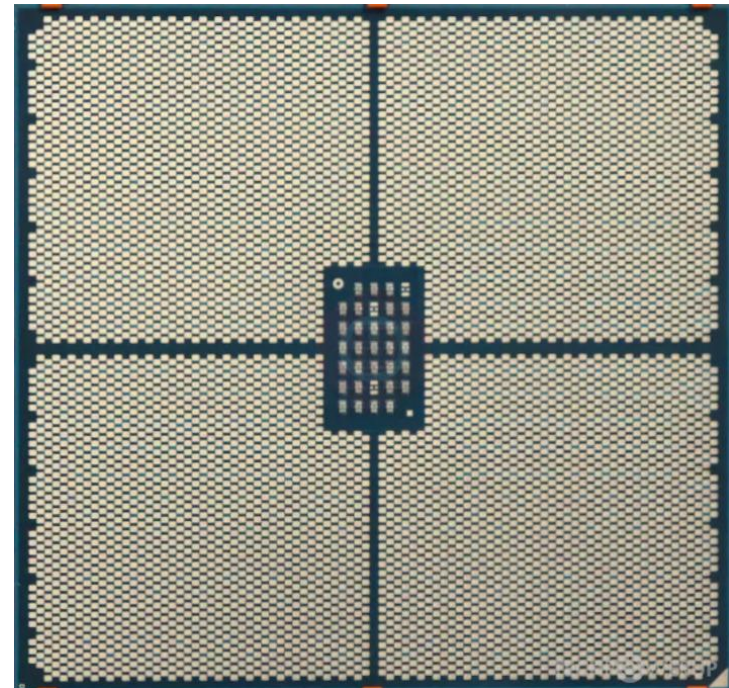
Very large scale integrated
circuits (VLSI) Computers

First microprocessor: Intel 4004 (transistor-based) in 1971



2250 Transistors / 10 micro meter

AMD Epyc 128cores/256threads in 2023

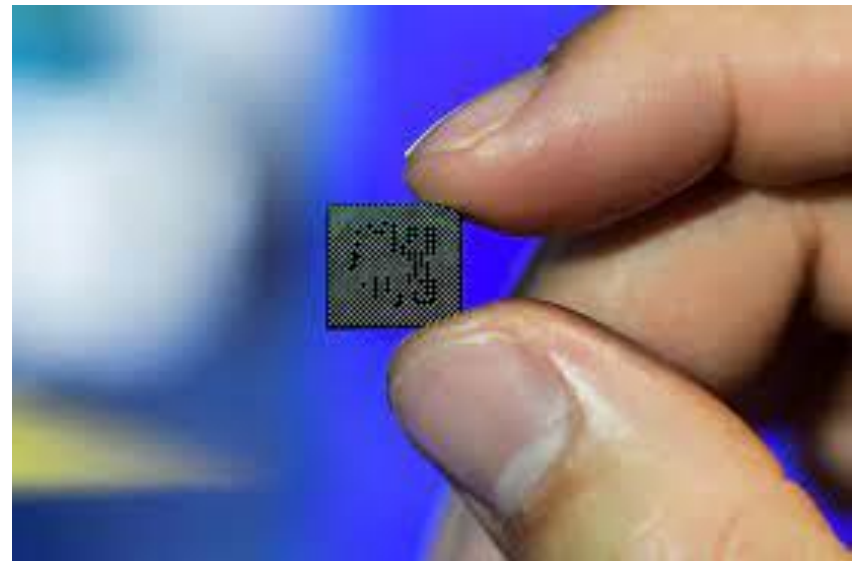


71,000,000,000 Transistors / 5 nanometer

Apple A17 pro, Snapdragon 8 Gen4



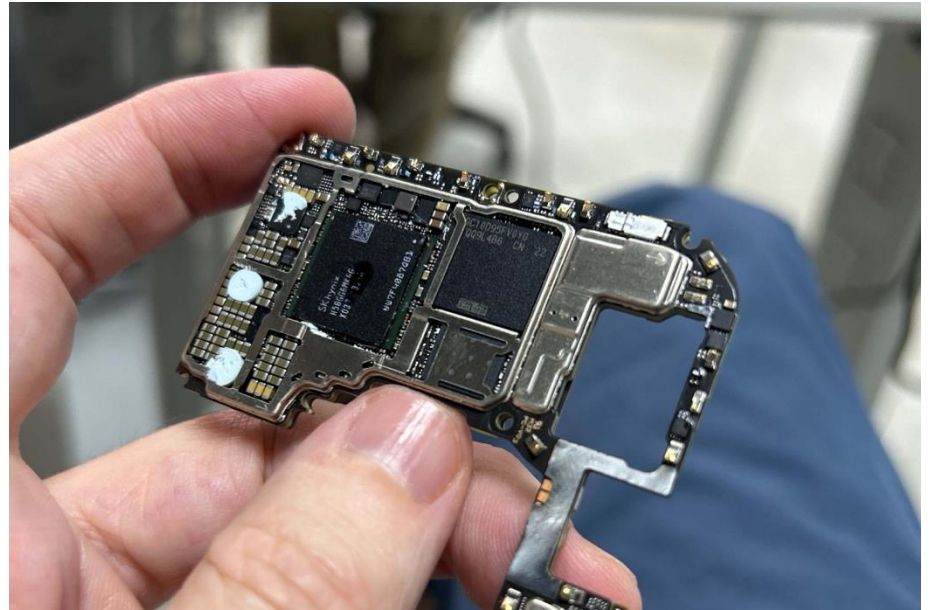
Apple Inc.



Digital Trends

19,000,000,000 Transistors / 3 nanometer

国产处理器龙芯3A6000， 麒麟9010



12 nanometer / 7 nanometer

Our World
in Data

50,000,000,000



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The classification of Computers

- Specific
 - 针对性强、特定服务、专门设计（飞船、火箭）
- General（科学计算、数据处理、过程控制解决各类问题）
 - 超级计算机
 - 大型机
 - 服务器
 - 工作站
 - 微型机
 - 单片机

基本区别通常在于其体积大小、结构复杂程度、功率消耗、性能指标、数据存储容量、指令系统和设备、软件配置等的不同

Supercomputer 超级计算机

- 运算速度很高，可达每秒执行几亿条指令，数据存储容量很大，规模大结构复杂，价格昂贵，主要用于大型科学计算。它也是衡量一国科学实力的重要标志之一。

Rank	System	Cores	Rmax (TFlop/s)	Rpeak (TFlop/s)	Power (kW)
1	Summit - IBM Power System AC922, IBM POWER9 22C 3.07GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband , IBM DOE/SC/Oak Ridge National Laboratory United States	2,282,544	122,300.0	187,659.3	8,806
2	Sunway TaihuLight - Sunway MPP, Sunway SW26010 260C 1.45GHz, Sunway , NRCPC National Supercomputing Center in Wuxi China	10,649,600	93,014.6	125,435.9	15,371
3	Sierra - IBM Power System S922LC, IBM POWER9 22C 3.1GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband , IBM DOE/NNSA/LLNL United States	1,572,480	71,610.0	119,193.6	
4	Tianhe-2A - TH-IVB-FEP Cluster, Intel Xeon E5-2692v2 12C 2.2GHz, TH Express-2, Matrix-2000 , NUDT National Super Computer Center in Guangzhou China	4,981,760	61,444.5	100,678.7	18,482
5	AI Bridging Cloud Infrastructure (ABCI) - PRIMERGY CX2550 M4, Xeon Gold 6148 20C 2.4GHz, NVIDIA Tesla V100 SXM2, Infiniband EDR , Fujitsu National Institute of Advanced Industrial Science and Technology (AIST) Japan	391,680	19,880.0	32,576.6	1,649
6	Piz Daint - Cray XC50, Xeon E5-2690v3 12C 2.6GHz, Aries interconnect , NVIDIA Tesla P100 , Cray Inc. Swiss National Supercomputing Centre (CSCS) Switzerland	361,760	19,590.0	25,326.3	2,272

太湖之光



大型机 Mainframe

- 大型机使用专用的处理器指令集、操作系统和应用软件。
- 从[IBM System/360](#)开始的一系列计算机及与其兼容或同等级的计算机，主要用于大量数据和关键项目的计算，例如银行金融交易及数据处理、人口普查、企业资源规划等等。
- RAS(Reliability, Availability, Serviceability 高可靠性、高可用性、高服务性)



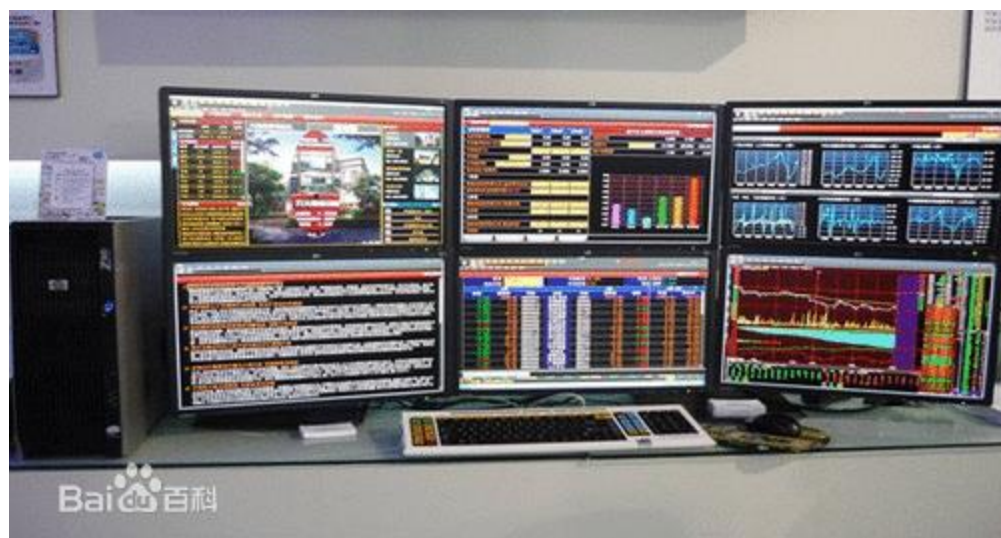
服务器 Server

- **服务器**（Server）指一个管理资源并为用户提供服务的计算机，通常分为文件服务器、数据库服务器和应用程序服务器。运行以上软件的计算机或计算机系统也被称为服务器。相对于普通PC来说，服务器在稳定性、安全性、性能等方面都要求更高，因此硬件和普通PC有所不同



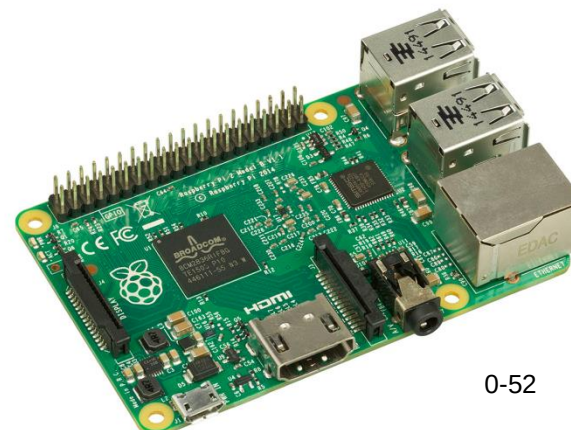
工作站 Workstation

- 工作站是一种高端的通用微型计算机。它是为了单用户使用并提供比个人计算机更强大的性能，尤其是在图形处理能力，任务并行方面的能力。通常配有高分辨率的大屏、多屏显示器及容量很大的内存储器 and 外部存储器，并且具有极强的信息和高性能的图形、图像处理功能的计算机。
另外，连接到服务器的终端机也可称为工作站



微型机及单片机

- 微型机
 - 台式机、笔记本电脑、Pad、Phone
- 单片机
 - 单片机 (Single chip microcomputer) 是一种集成电路芯片，把具有数据处理能力的中央处理器CPU、存储器等集成到一块硅片上构成的一个小而完善的微型计算机系统，在工业控制领域广泛应用。
 - 片上系统SoC (system on chip)



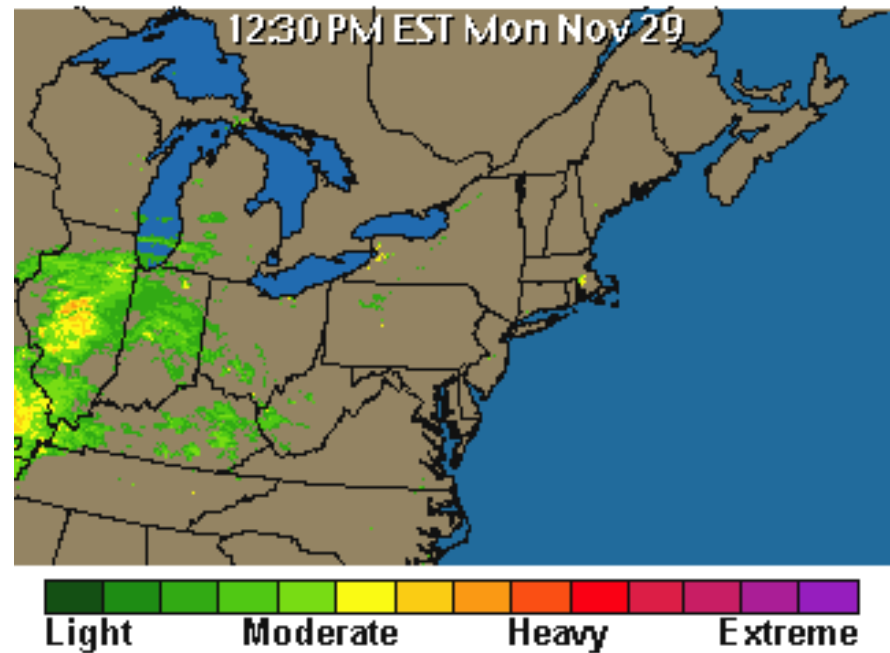
What can computers do?

- Numeric computing
- Information processing
- Real-time control
- Computer aided design
- Artificial intelligence



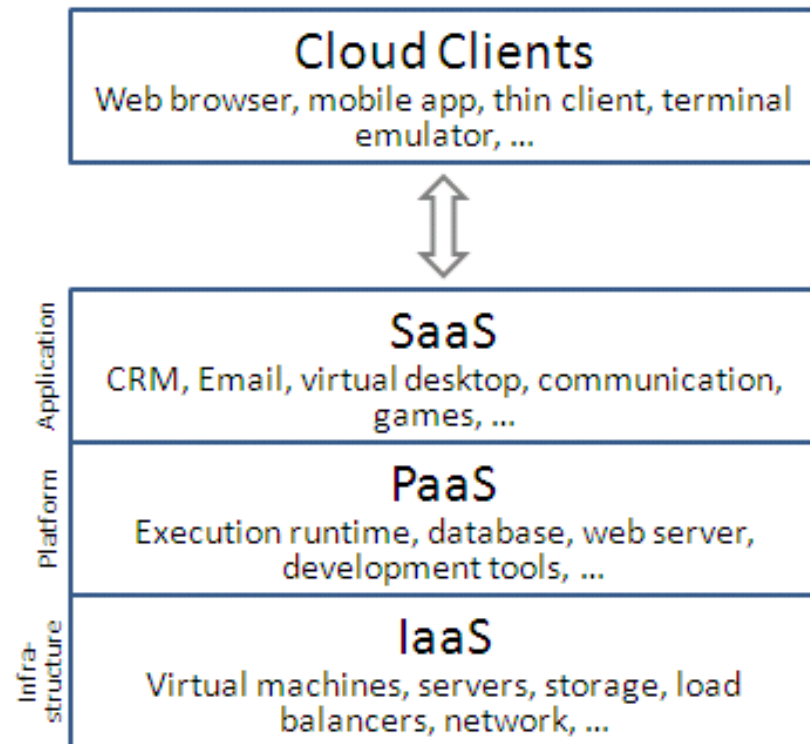
Super computing

- Climate prediction & Weather forecasting
- Computational chemistry
- Crash analysis
- Cryptography
- Nuclear simulation
- Structural analysis



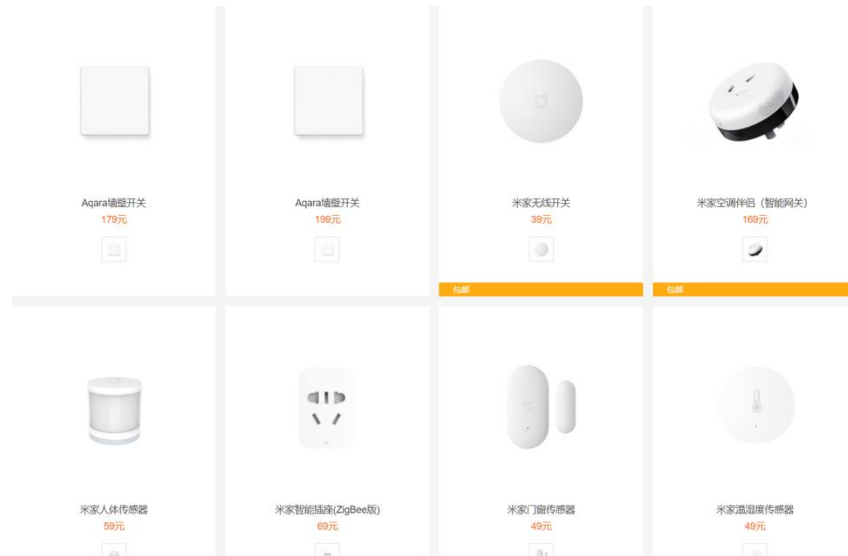
云计算 Cloud Computing

- 云计算是一种按使用量付费的模式,这种模式提供可用的、便捷的、按需的网络访问,进入可配置的计算资源共享池
 - Public Cloud
 - Private Cloud
 - Hybrid Cloud



物联网 Internet of things (IoTs)

- 物联网的核心和基础仍然是互联网，是在互联网基础上的延伸和扩展的网络；物联网就是“物物相连的互联网”。物联网通过智能感知、识别技术与普适计算、泛在网络的融合应用，被称为继计算机、互联网之后世界信息产业发展的第三次浪潮。



大数据

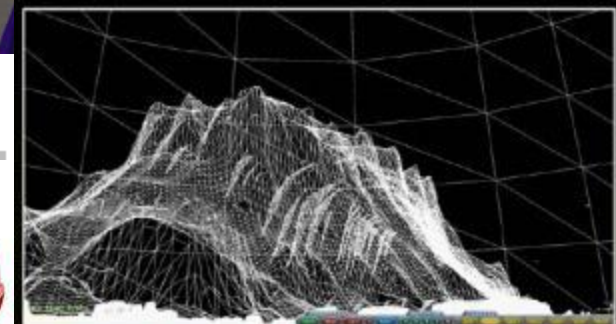
- 大数据(big data), 或称巨量资料, 指的是所涉及的资料量规模巨大到无法透过目前主流软件工具, 在合理时间内达到撷取、管理、处理、并整理成为帮助企业经营决策更积极目的的资讯
- **技术**: 大规模并行处理 (MPP) 数据库、数据挖掘电网、分布式文件系统、分布式数据库、云计算平台、互联网和可扩展的存储系统
- **例子**: 网络日志, RFID, 传感器网络, 社会网络, 社会数据 (由于数据革命的社会), 互联网文本和文件;互联网搜索索引;呼叫详细记录, 天文学, 大气科学, 基因组学, 生物地球化学, 生物, 和其他复杂和/或跨学科的科研, 军事侦察, 医疗记录;摄影档案馆视频档案;和大规模的电子商务
 - 芝麻信用
 - 随申码

信息安全技术及应用

- 有网络安全、主机（服务器）安全、应用（系统）安全、数据安全，以及桌面终端安全
- 计算机信息安全技术 培养掌握系统与网络安全的基本理论与病毒防范、黑客攻击手段分析与防范技术，能熟练应用信息安全产品，熟悉信息安全管理规范，具有开发、维护和管理信息安全系统能力的高等技术应用性人才。



- 虚拟现实，计算机图形学



3D立体显示 AR



- Video

Artificial Intelligence 人工智能

人工智能(Artificial Intelligence)，英文缩写为AI。它是研究、开发用于模拟、延伸和扩展人的智能的理论、方法、技术及应用系统的一门新的技术科学。人工智能是计算机科学的一个分支，它企图了解智能的实质，并生产出一种新的能以人类智能相似的方式作出反应的智能机器



Computer vision



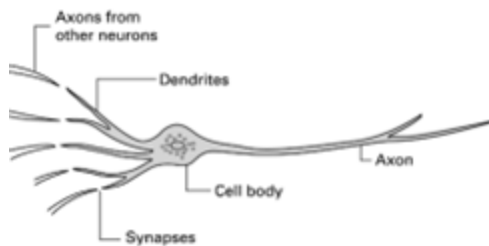
Natural language



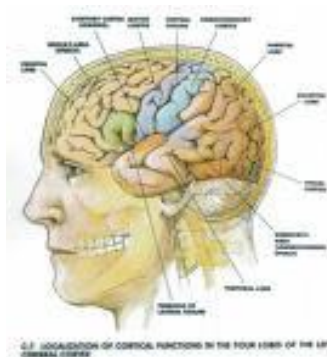
Games



Robot



Neural network

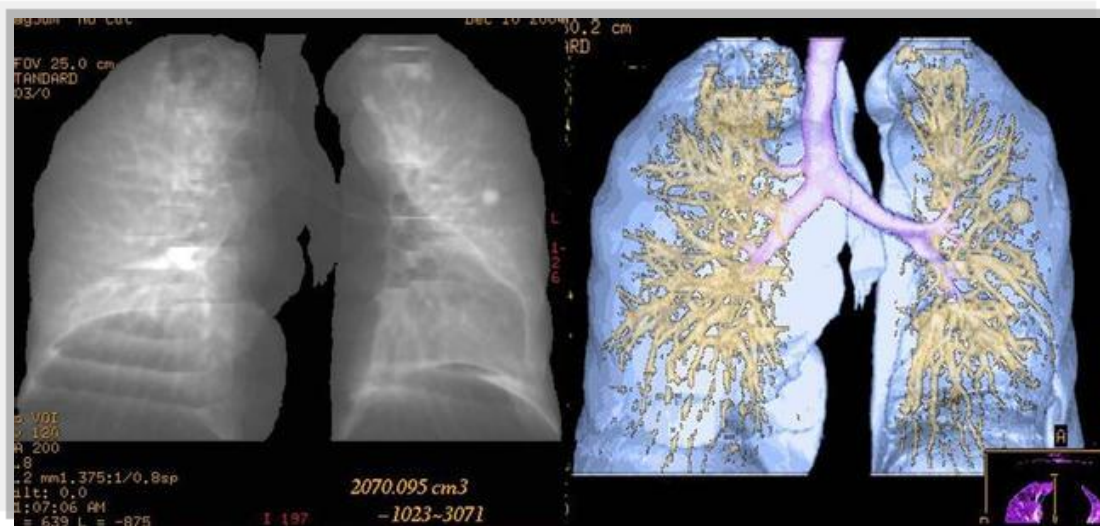
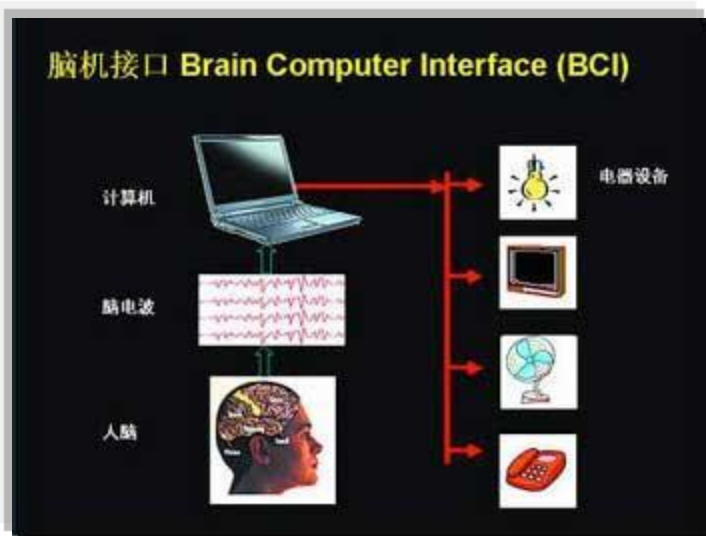


Perception



Data mining

机器视觉，医学图像处理



• 机器学习

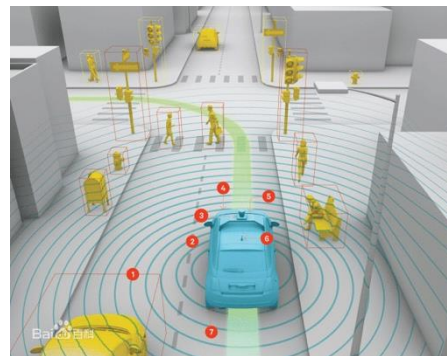
- 机器学习(Machine Learning)是一门多领域交叉学科，涉及概率论、统计学、逼近论、凸分析、算法复杂度理论等多门学科。专门研究计算机怎样模拟或实现人类的学习行为，以获取新的知识或技能，重新组织已有的知识结构使之不断改善自身的性能
- 应用：网页聚类分类，相关性排序，广告点击率预测，用户分类，精准广告，金融地产数据分析，医学数据分析



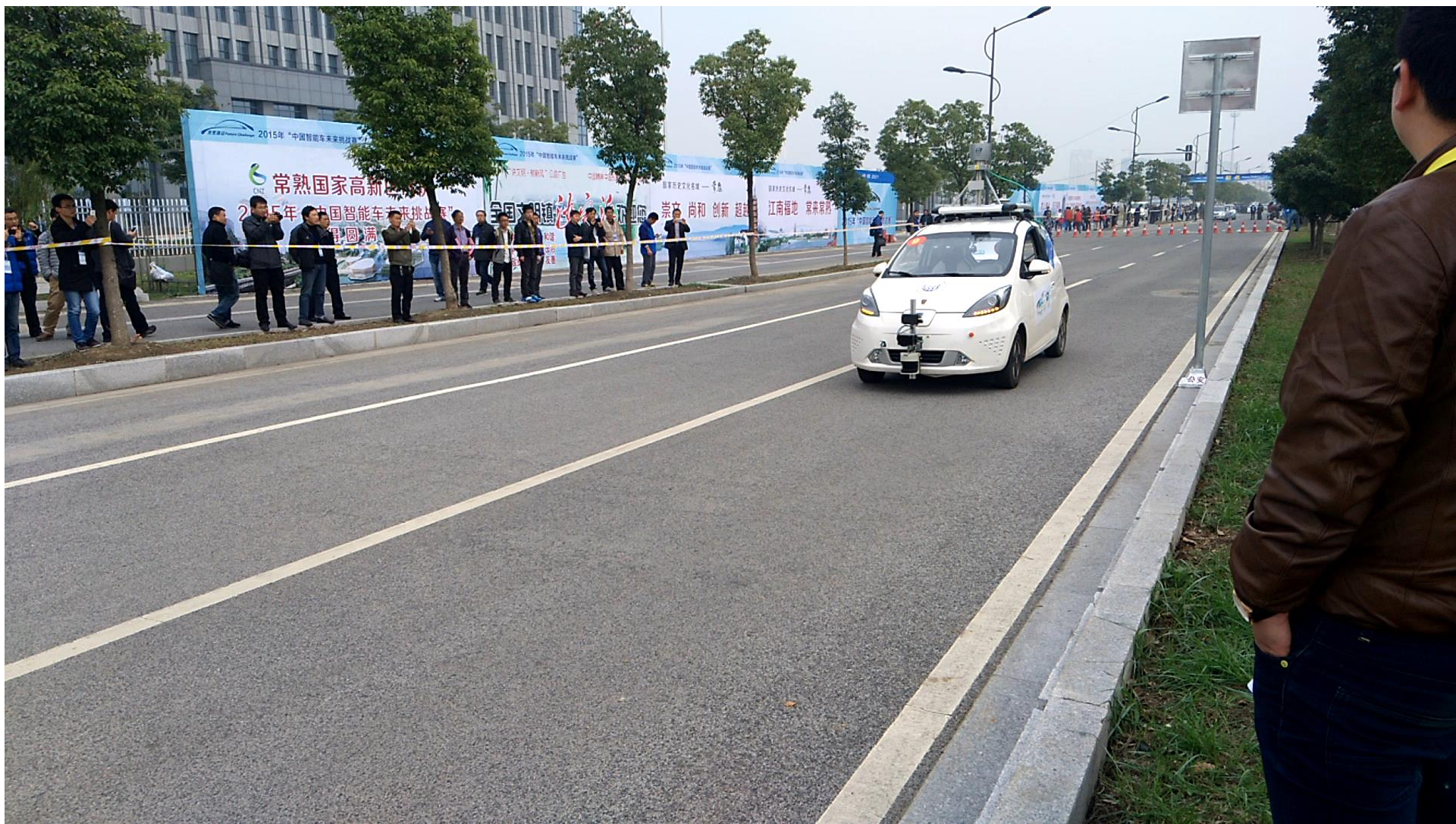
- Video

无人驾驶汽车

- 无人驾驶汽车是通过车载传感系统感知道路环境，自动规划行车路线并控制车辆到达预定目标的智能汽车



TiEV 途灵



- Questions?

- **Let's play a game!**

- Step1: Please stand up
- Step2: Each student initiate with number $X = 1$
- Step 3: Add the X of your closest standing neighbor to your X $X_{\text{new}} = X_{\text{old}} + X$
- Step 4: Please let your neighbor **sit down**
- The remaining standing students repeat from Step 3 to Step 4